NETFLIX CONTENT STRATEGY & AUDIENCE CLASSIFICATION

Course End Project submitted in partial fulfillment of the Requirements for the Award of the Degree of

BACHELOR OF ENGINEERING

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

APPLIED PREDICTIVE ANALYTICS (22ADE57N)

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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CERTIFICATE

This is to certify that the course end project entitled “Netflix Content Strategy & Audience Classification” being submitted by

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in partial fulfillment for the award of the Degree of Bachelor of Engineering in Artificial Intelligence and Data Science to the CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY, HYDERABAD, carried out under my mentorship and supervision during the year 2026-27. The results presented in this project have not been submitted to any other university or institute for the award of any degree.

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ABSTRACT

Netflix operates a large catalogue of movies and television programmes originating from different countries, languages and production environments. Organising this catalogue by format, genre, production country and intended audience is important for content discovery, parental controls and catalogue planning. This project, titled “Netflix Content Strategy & Audience Classification”, analyses the public Netflix Movies and TV Shows dataset obtained from Kaggle (shivamb/netflix-shows).

The study combines exploratory data analysis with a three-class audience-group classification task. The first part examines the distribution of Movies and TV Shows, yearly additions, leading genres, audience categories and content-producing countries. The second part uses metadata such as content type, release year, duration, genre count and primary production country to predict the audience group derived from official content ratings.

Two supervised learning algorithms are considered: Multinomial Logistic Regression and Random Forest. The supplied evaluation outputs show that Random Forest achieved an accuracy of approximately 60.76%, compared with 56.82% for Logistic Regression. The results indicate that metadata contains useful predictive information, although the confusion matrices also show substantial confusion between audience groups. The report presents the dataset, preparation steps, visual evidence, model evaluation, findings, limitations and possible improvements.

Keywords: Netflix, exploratory data analysis, audience classification, Logistic Regression, Random Forest, metadata, predictive analytics.

PROJECT ORGANISATION

Section	Contents
1	Introduction, business context and objectives
2	Dataset source, structure and data dictionary
3	Data preparation and feature engineering
4	Exploratory analysis and five visualizations
5	Predictive modelling and evaluation
6	Key findings, implications and recommendations
7	Limitations, future scope and conclusion
8	Tools, reproducibility and references

1. INTRODUCTION

1.1 Background

Digital streaming platforms provide large and continuously changing libraries of films and television programmes. A catalogue may contain thousands of titles, each described by metadata such as content type, release year, country, genre, duration and official rating. These attributes help users discover relevant content and help platform teams organise the catalogue.

Netflix is a suitable example because its public catalogue dataset contains a mixture of numerical, categorical and text-derived attributes. Studying these fields makes it possible to understand the composition of the catalogue and to explore whether a title's broad audience category can be estimated from its metadata.

1.2 Business Problem

Audience labels are important for age-appropriate discovery, parental controls and recommendation systems. However, content ratings are represented through multiple rating codes, and catalogues may combine metadata supplied by different distributors and regions. Manually reviewing every title can be time-consuming. A predictive model can provide an initial classification or flag titles for further review.

The central question addressed in this project is: Can a title's audience group be predicted from the available catalogue metadata?

1.3 Project Objectives

- Explore the Netflix catalogue and identify patterns in content format, additions over time, genres and production countries.

- Convert raw metadata into clear visualizations that support interpretation and discussion.
- Create three broad audience classes—Kids/Family, Teens and Adults—from the available rating codes.
- Train and evaluate Logistic Regression and Random Forest classifiers using structured metadata.
- Interpret the results and identify practical implications, limitations and future improvements.

1.4 Scope

The work focuses on the supplied public catalogue snapshot and the fields available in netflix_titles.csv. It does not evaluate individual titles using their video or audio content, nor does it measure viewer satisfaction, watch time, revenue or subscription impact. The model predicts the engineered audience label; it is not a replacement for official rating authorities or human review.

2. DATASET DESCRIPTION

2.1 Dataset Source

The project uses the “Netflix Movies and TV Shows” dataset published on Kaggle under the identifier shivamb/netflix-shows. The file used is netflix_titles.csv. The dataset contains approximately 8,800 title records and 12 columns and represents a public catalogue snapshot from around 2021.

Attribute	Details
Dataset name	Netflix Movies and TV Shows
Platform / publisher	Kaggle — shivamb
Dataset identifier	shivamb/netflix-shows
File used	netflix_titles.csv
Approximate size	About 8,800 rows and 12 columns
Data nature	Public catalogue metadata; snapshot around 2021
Access method	kagglehub download in Google Colab; manual CSV upload is a fallback

2.2 Dataset Attributes

The raw dataset includes identifiers, title information, production details, date information, content ratings, duration, genre tags and a short description. The table below summarises how each field is treated in this project.

Column	Description	Project use
show_id	Unique title identifier	Not used as a model feature
type	Movie or TV Show	Charts; encoded model feature
title	Title name	Not used
director	Director or directors	Not used; many missing values
cast	Cast members	Not used
country	Production country/countries	Primary country for chart/model
date_added	Date added to Netflix	Extract year_added for growth chart
release_year	Original release year	Numeric model feature
rating	Official content rating	Source of audience-group label
duration	Minutes or seasons	Parsed numeric model feature
listed_in	Genre tags	Genre chart and num_genres
description	Short synopsis	Not used in current model

2.3 Data Quality Observations

- The director field is missing for a substantial share of titles; cast and country also contain missing values.
- date_added, rating and duration contain some missing entries; rows without required modelling fields are removed during preparation.
- The duration field combines minutes for movies and seasons for TV shows, so its numeric component must be extracted.
- Country and genre fields may contain multiple comma-separated values in one cell.

- Non-standard or unsupported rating values cannot be mapped reliably to the three audience classes and are left without a target label.

3. DATA PREPARATION AND FEATURE ENGINEERING

3.1 Data Cleaning and Parsing

The raw catalogue is transformed into a structured table suitable for descriptive analysis and classification. The notebook performs the following preparation steps:

Step	Operation	Purpose
1	Convert date_added to datetime and extract year_added	Supports yearly catalogue-addition analysis
2	Extract the numeric part of duration into duration_value	Creates a numeric representation of minutes or seasons
3	Take the first country listed as primary_country; missing values become Unknown	Provides one country category per title
4	Count comma-separated genre tags as num_genres	Represents the breadth of genre tagging
5	Map supported rating codes to audience_group	Creates the three-class prediction target
6	Drop rows missing audience_group, duration_value or year_added	Keeps rows with required analysis/model fields

3.2 Engineered Features

Feature	Source field	Meaning
year_added	date_added	Calendar year the title was added
duration_value	duration	Numeric duration: minutes for movies or seasons for TV shows
primary_country	country	First production country listed; Unknown if absent
num_genres	listed_in	Number of genre tags associated with the title
audience_group	rating	Target label: Kids/Family, Teens or Adults

3.3 Rating-to-Audience Mapping Used in the Notebook

The notebook maps selected rating codes into three broad groups. This is an engineered grouping for this project and should not be interpreted as a universal equivalence between rating systems.

Audience group	Rating codes mapped in the notebook
Kids/Family	G, PG, TV-Y, TV-Y7, TV-Y7-FV, TV-G
Teens	PG-13, TV-14
Adults	R, TV-MA, NC-17
Not mapped	TV-PG, NR, UR and malformed/other values

Important implementation note: the notebook's actual rating_map does not include TV-PG, although an earlier draft description listed it under Teens. This report follows the notebook implementation so the written method matches the executed code. Unmapped values do not receive an audience_group label and are excluded from the cleaned modelling table.

3.4 Modelling Feature Preparation

The type field is label-encoded. The primary_country field is grouped so that the 15 most frequent countries remain separate and all remaining countries are assigned to Other; the grouped field is then one-hot encoded. Numeric inputs are release_year, duration_value and num_genres. The target audience_group is label-encoded for model training.

4. EXPLORATORY DATA ANALYSIS AND VISUALIZATIONS

Five visualizations were prepared using pandas, seaborn and matplotlib. They describe the catalogue from complementary perspectives: format, additions over time, genre mix, audience distribution and production geography.

4.1 Visualization 1 — Content Type Split

The count chart shows 6,131 Movies and 2,676 TV Shows, for a combined total of 8,807 titles in the displayed catalogue. The proportion chart reports approximately 69.6% Movies and 30.4% TV Shows. The catalogue is therefore film-heavy, with television series representing just under one-third of the records.

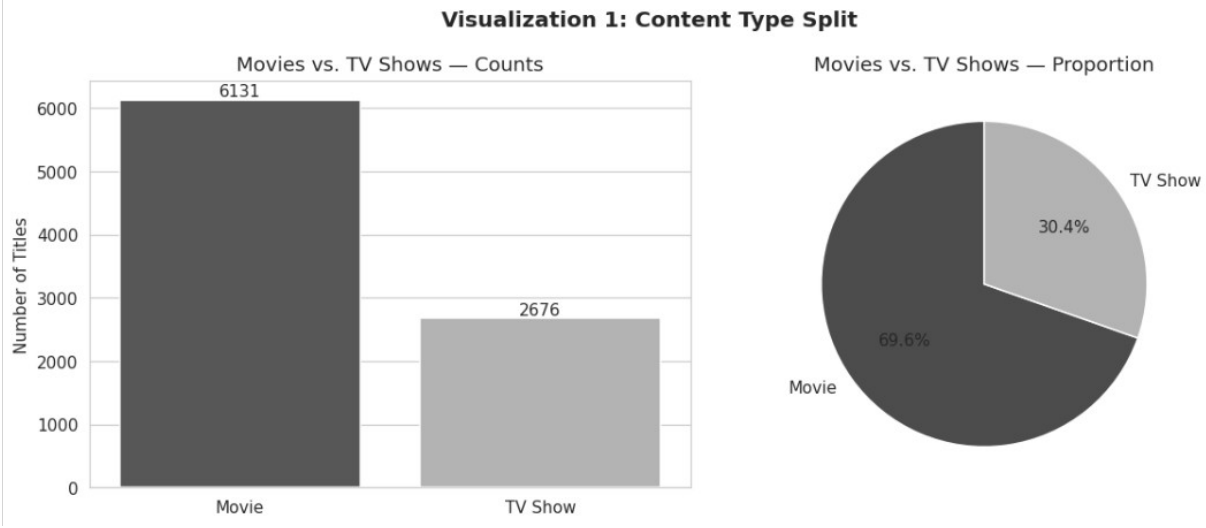


Figure 1. Movie and TV Show counts and their share of the catalogue.

Interpretation

The difference in format mix suggests that films form the larger part of the observed library. TV Shows remain a meaningful segment and may support episodic viewing and longer engagement, although engagement itself is not measured in this dataset.

4.2 Visualization 2 — Titles Added to Netflix per Year

The stacked area chart separates yearly additions into Movies and TV Shows. The visible pattern shows limited additions in the early years, followed by a sharp rise from approximately 2015. The highest combined addition level occurs around 2019, after which the yearly totals decline in 2020 and 2021.

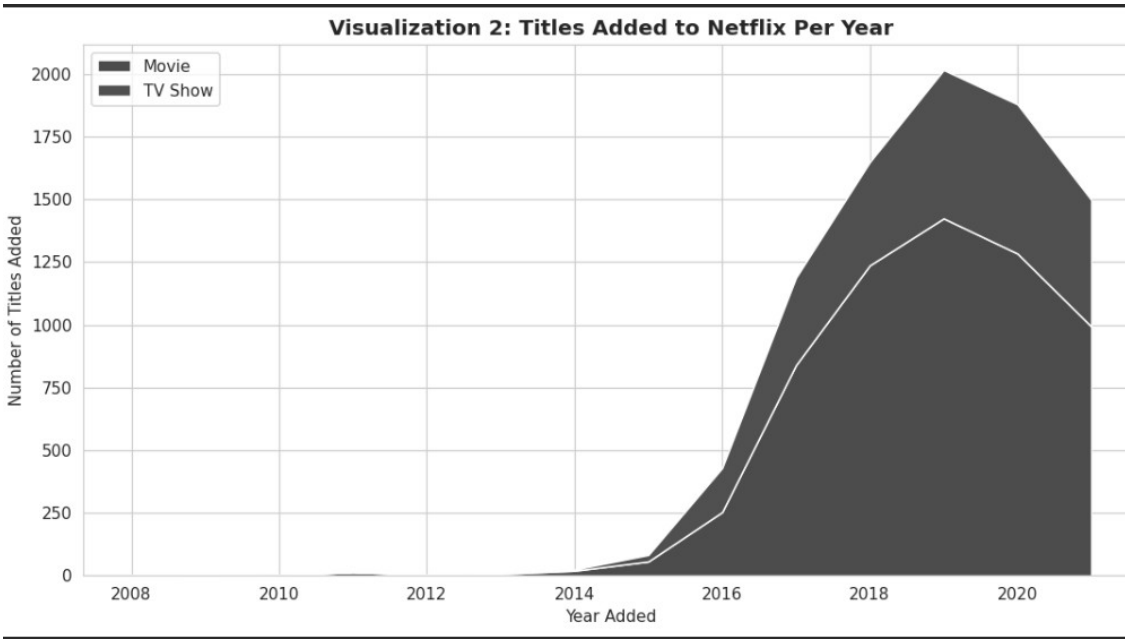


Figure 2. Yearly additions to Netflix, split by Movies and TV Shows.

Interpretation

The growth pattern indicates a period of rapid catalogue expansion followed by a slower pace in the final years shown. The chart describes the dates at which titles were added to the dataset; it does not independently establish why additions increased or decreased.

4.3 Visualization 3 — Top 10 Genres

The horizontal bar chart ranks the most frequent genre labels. International Movies is the leading category, followed by Dramas and Comedies. International TV Shows and Documentaries are also prominent, while Action & Adventure, TV Dramas, Independent Movies, Children & Family Movies and Romantic Movies complete the top ten.

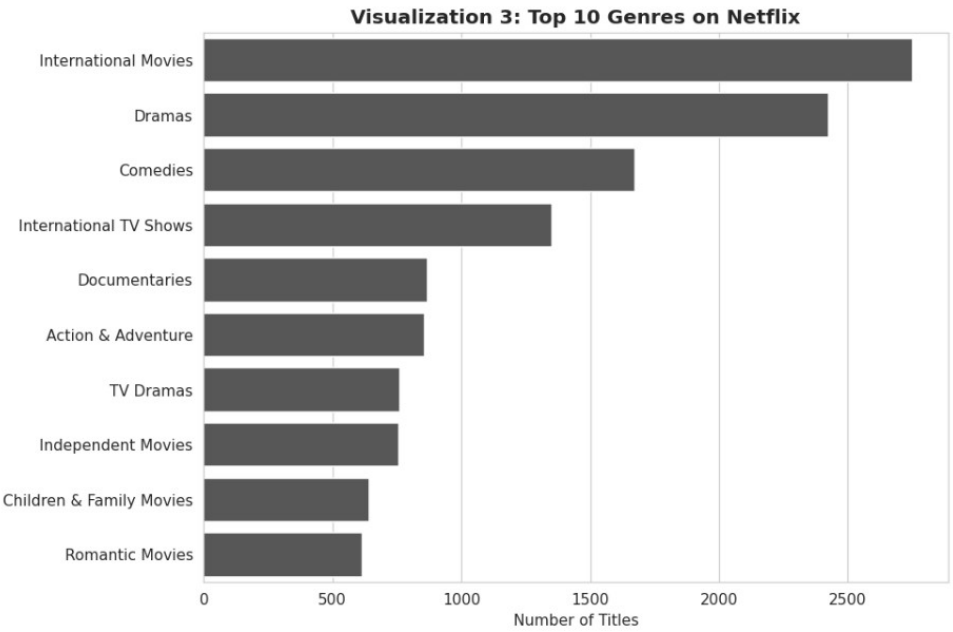


Figure 3. Ten most frequent genre categories in the Netflix catalogue.

Interpretation

The leading categories indicate a catalogue with substantial international, dramatic and comedy content. Because a title can carry more than one genre tag, genre counts represent tag frequency rather than mutually exclusive title totals.

4.4 Visualization 4 — Content Distribution by Audience Group

The audience-group chart contains 1,193 Kids/Family titles, 2,647 Teens titles and 4,007 Adults titles. Adults form the largest group, representing roughly half of the displayed distribution. Kids/Family is the smallest group.

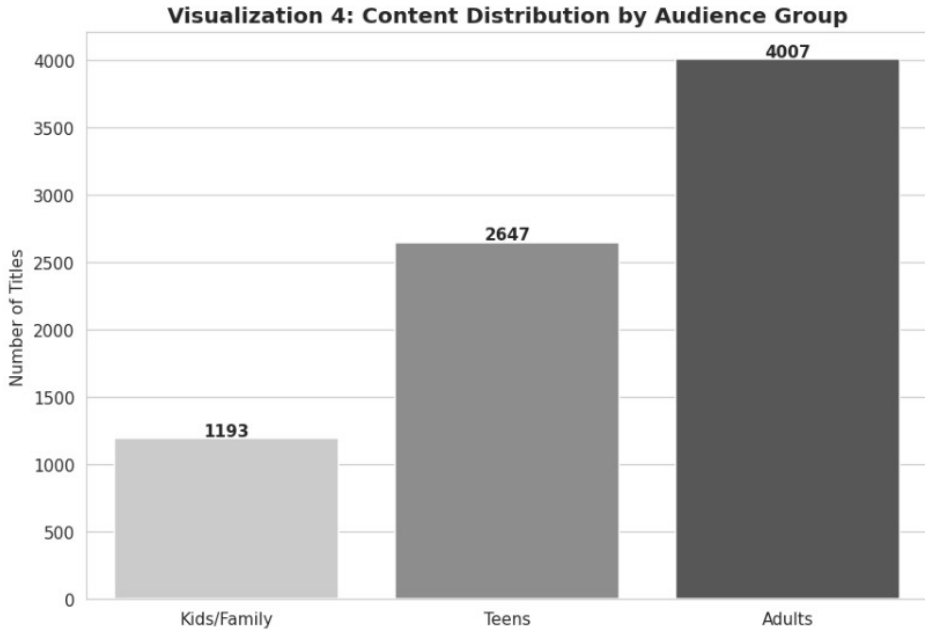


Figure 4. Distribution of titles across the three engineered audience groups.

Interpretation

The distribution is imbalanced, with more Adult-labelled records than Kids/Family records. This imbalance is relevant when evaluating a classifier: overall accuracy alone can hide weaker performance on the smaller class. Macro-averaged precision, recall and F1 are therefore useful complementary measures.

4.5 Visualization 5 — Top 10 Content-Producing Countries

The country chart shows the United States as the largest contributor, followed by India and the United Kingdom. Unknown country values also appear among the top categories, reflecting missing country metadata that has been explicitly retained as an Unknown category. Canada, Japan, France, South Korea, Spain and Mexico complete the displayed top ten.

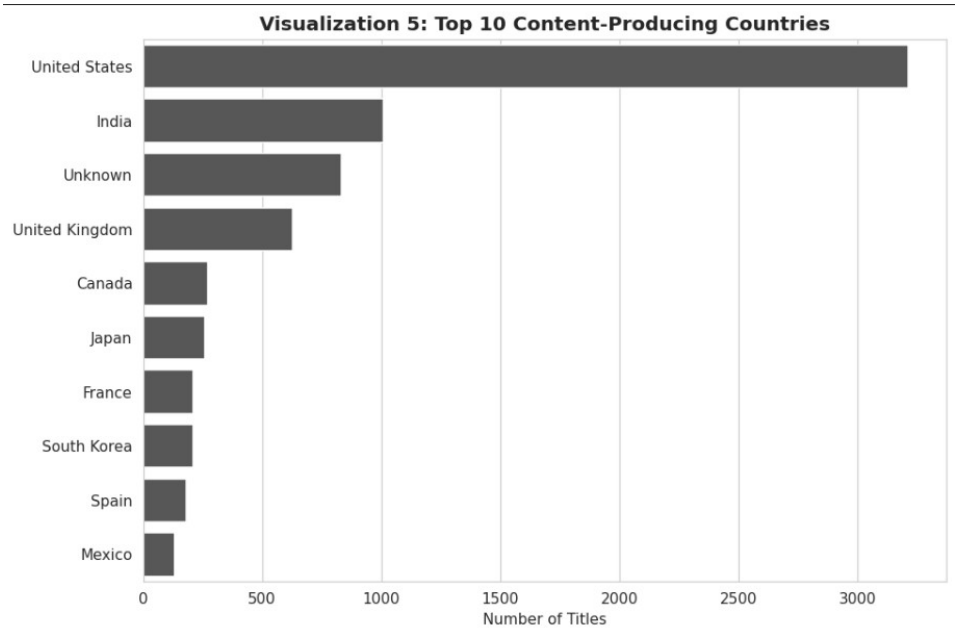


Figure 5. Top ten primary-country categories by title count.

Interpretation

The catalogue has a strong US contribution alongside meaningful representation from India and other production markets. The Unknown category should be interpreted as incomplete metadata, not as a real production country.

5. PREDICTIVE MODELLING

5.1 Classification Task

The second part of the project predicts the engineered audience_group label from title metadata. This is a supervised, three-class classification problem with the classes Kids/Family, Teens and Adults.

5.2 Input Features and Target

Component	Configuration used
Target variable	audience_group: Kids/Family, Teens, Adults
Numeric features	release_year, duration_value, num_genres
Categorical features	type (label encoded); grouped primary_country (one-hot encoded)
Country grouping	Top 15 primary countries retained; remaining values grouped as Other
Data split	80% training and 20% testing
Split strategy	Stratified by class; random_state = 42

5.3 Algorithm 1 — Multinomial Logistic Regression

Multinomial Logistic Regression is used as a baseline classifier for the three audience groups. It estimates class probabilities from a weighted combination of input features and assigns the class with the strongest estimated probability. The notebook sets max_iter=2000 and random_state=42. Logistic Regression is relatively straightforward to interpret and provides a baseline against which a more flexible model can be compared.

5.4 Algorithm 2 — Random Forest Classifier

Random Forest is an ensemble learning method that combines predictions from multiple decision trees. It can represent non-linear relationships and interactions between metadata fields. The notebook uses 300 trees, max_depth=12, random_state=42 and n_jobs=-1. The fitted model also provides feature-importance values, which are used to identify influential input variables.

5.5 Evaluation Measures

Measure	Meaning in this study
Accuracy	Proportion of all test records classified correctly
Macro precision	Precision calculated for each class and averaged equally
Macro recall	Recall calculated for each class and averaged equally
Macro F1-score	Harmonic mean of precision and recall per class, averaged equally
Confusion matrix	Counts correct and incorrect predictions for each actual/predicted class pair

6. MODEL RESULTS AND EVALUATION

6.1 Confusion Matrix Results

The supplied confusion-matrix output compares actual and predicted labels for 1,570 test records. The class order in the matrices is Adults, Kids/Family and Teens. Correct predictions lie on the main diagonal; off-diagonal values show where the model confuses one audience group with another.

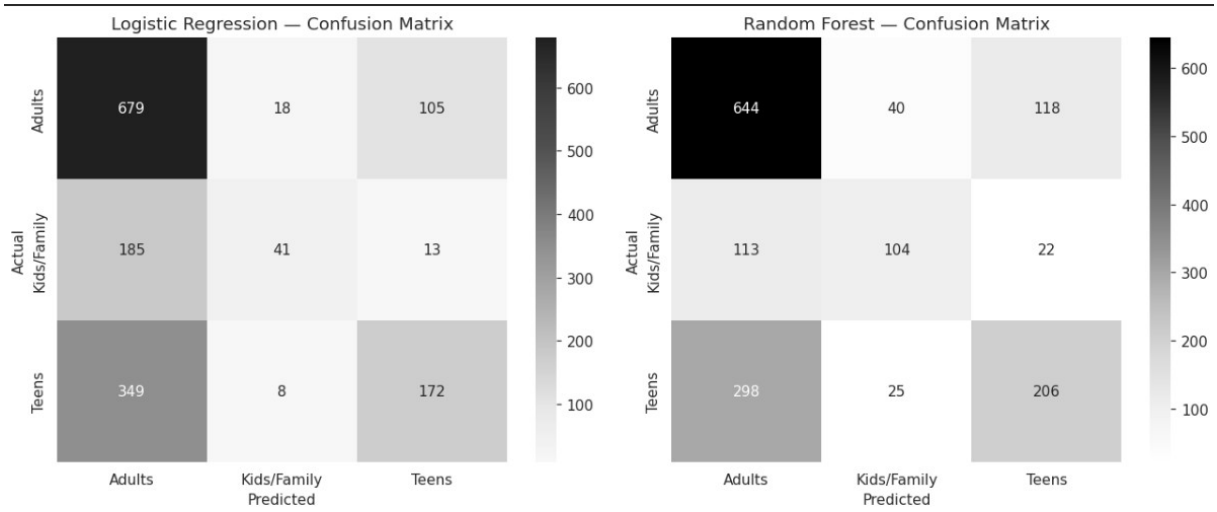


Figure 6. Confusion matrices for Logistic Regression and Random Forest.

6.2 Metrics Calculated from the Displayed Confusion Matrices

The following metrics are calculated from the cell counts visible in Figure 6. Values are rounded to three decimal places.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Logistic Regression	0.568	0.588	0.448	0.454
Random Forest	0.608	0.607	0.543	0.558

6.3 Result Interpretation

Random Forest performs better than Logistic Regression on all four displayed aggregate measures. Its accuracy is approximately 60.8%, compared with 56.8% for Logistic Regression. Its macro F1-score is also higher (0.558 versus 0.454), suggesting a better balance of precision and recall across the three classes.

Both models show difficulty distinguishing the audience groups. In particular, many actual Teens titles are predicted as Adults, and many actual Adults titles are also predicted as Teens. Kids/Family has fewer correct predictions than the Adult class in both matrices, although Random Forest improves the Kids/Family diagonal count from 41 to 104.

The confusion matrices therefore support a cautious conclusion: the available metadata provides some predictive signal, and Random Forest is the stronger of the two tested models, but neither model should be treated as sufficiently reliable for automatic age-rating decisions without further validation.

7. FEATURE IMPORTANCE AND KEY FINDINGS

7.1 Random Forest Feature Importance

The Random Forest feature-importance chart ranks the variables by their contribution to the fitted ensemble. Duration_value has the highest displayed importance, followed by release_year and num_genres. Country indicators, including country_India and country_Unknown, contribute smaller amounts.

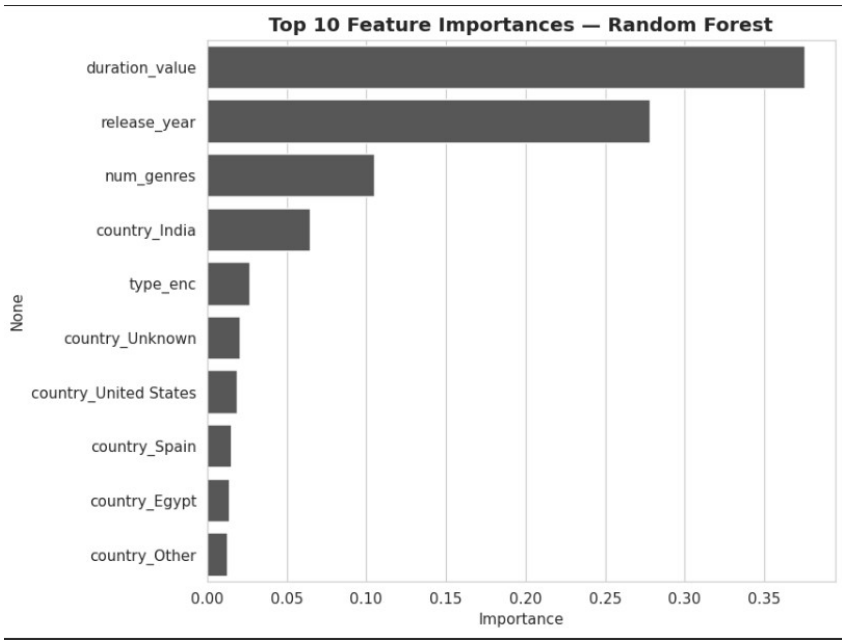


Figure 7. Top ten feature-importance values reported by Random Forest.

7.2 Interpretation of Important Features

- duration_value is the strongest feature in the displayed ranking. Runtime or season count is associated with patterns learned by the model, but this does not mean duration causes a particular audience rating.
- release_year is the second most influential feature, indicating that the title's original release period contributes to the model's predictions.
- num_genres captures how many genre labels are attached to a title and provides additional metadata signal.
- Country indicators have lower individual importance, but production geography still contributes to the model when combined with other features.

7.3 Consolidated Findings

Area	Finding from the supplied outputs
Content format	6,131 Movies (69.6%) and 2,676 TV Shows (30.4%).
Catalog additions	Additions rise sharply from about 2015, peak around 2019 and then decline in the final years shown.
Genres	International Movies, Dramas and Comedies are the three leading genre categories.
Audience mix	Adults (4,007) exceed Teens (2,647) and Kids/Family (1,193).
Countries	United States leads; India and United Kingdom follow. Unknown is a metadata category.
Model comparison	Random Forest outperforms Logistic Regression on the displayed aggregate metrics.
Important predictors	duration_value, release_year and num_genres lead the feature-importance ranking.

8. BUSINESS IMPLICATIONS AND RECOMMENDATIONS

8.1 Catalogue Planning

The movie-heavy distribution and the strong presence of international movies, dramas and comedies provide a high-level view of the catalogue mix. Content teams could use this type of analysis to monitor whether the mix remains aligned with regional demand and the platform's programming objectives. The dataset does not contain viewing or revenue measures, so the chart findings should be used as catalogue-composition evidence rather than proof of commercial success.

8.2 Audience-Aware Discovery

A three-group audience label can support broad catalogue filtering and quality-control workflows. A model prediction could be used as a review flag when metadata appears inconsistent with an existing rating. Final age suitability decisions should remain based on official ratings and appropriate human review.

8.3 Regional Content Strategy

The United States is the largest production-country category, while India and the United Kingdom are also prominent. A regional strategy team could monitor country-level representation and investigate whether specific markets or languages are under-represented. Since some titles list multiple countries and the current analysis uses only the first listed country, the chart is a simplified view of production geography.

8.4 Model Improvement Recommendations

- Evaluate per-class precision, recall and F1 in addition to overall accuracy, with particular attention to the smaller Kids/Family class.
- Use cross-validation and hyperparameter tuning to check whether the observed Random Forest advantage is stable.
- Investigate class-balancing methods or class weights to reduce the effect of audience-group imbalance.
- Test gradient-boosting models and compare them with the current baseline and Random Forest.
- Add text features from title descriptions, such as TF-IDF, only after careful validation and leakage checks.
- Review the rating-to-audience mapping with the course instructor or a documented rating policy before using the classifier operationally.

9. LIMITATIONS AND FUTURE SCOPE

9.1 Limitations

- The dataset is a catalogue snapshot from around 2021 and does not represent the current Netflix catalogue.
- No viewership, watch-time, revenue, user-rating or engagement fields are available; the project cannot explain title popularity or business performance.
- The target audience_group is derived from official rating codes. The model learns to reproduce this engineered mapping rather than independently assess the content itself.
- The notebook maps only selected rating codes. TV-PG is not included in the implemented mapping, and unsupported values are excluded from the modelling data.
- The duration_value field combines minutes and number of seasons. These are different units, so a single numeric feature may be difficult to interpret consistently across Movies and TV Shows.
- Only structured metadata is used in the current model. The description field and other potentially informative text are not included.
- The first listed production country is used as the primary country, which simplifies multi-country productions.
- The confusion matrices show meaningful class confusion, so the reported model performance is not sufficient for high-stakes automated classification.

9.2 Future Enhancements

- Use a more complete and current dataset and record its retrieval date for reproducibility.
- Review and document the rating-to-audience mapping, including how TV-PG and regional rating systems should be treated.
- Represent movie runtime and TV-show seasons as separate features, or model Movies and TV Shows separately.
- Apply stratified cross-validation, hyperparameter search and per-class evaluation.
- Explore class weighting, resampling or other methods to improve performance on the smaller Kids/Family class.

- Evaluate gradient boosting and calibrated probability estimates.
- Add synopsis text features and, where legally and ethically appropriate, richer content metadata.
- Test the model on a later catalogue snapshot to assess whether performance generalises over time.

10. CONCLUSION

This project demonstrates how a public streaming catalogue can be analysed using exploratory data analysis and supervised machine learning. The descriptive visualizations show that the supplied Netflix snapshot contains substantially more Movies than TV Shows, that catalogue additions increased rapidly from around 2015 and peaked around 2019, and that International Movies, Dramas and Comedies are the most frequent genre categories in the displayed top ten. The audience-group distribution is dominated by Adults, while the United States is the largest primary-country category.

The predictive component formulates audience classification as a three-class problem using content type, release year, duration, genre count and grouped production-country features. Random Forest achieves approximately 60.8% accuracy and a macro F1-score of 0.558 based on the supplied confusion matrix, outperforming the Logistic Regression baseline. However, the confusion matrices show that audience groups are still frequently misclassified, especially between Adults and Teens.

The most influential Random Forest features are duration_value, release_year and num_genres. These results suggest that catalogue metadata contains useful patterns, but the model remains an exploratory decision-support approach rather than a replacement for official rating processes. Better label definitions, class-aware validation, richer features and testing on newer data are important next steps.

Overall, the project fulfils its objective of converting raw catalogue metadata into interpretable visual insights and a baseline audience-classification model. It also illustrates how predictive analytics can support content organisation and strategic review when its limitations are clearly understood.

11. TOOLS AND TECHNOLOGIES

Tool / Library	Purpose in the project
Google Colab	Cloud notebook environment for running the analysis
Python 3	Programming language for data processing and modelling
kagglehub	Retrieves the dataset from Kaggle
pandas	Loads, cleans, transforms and aggregates tabular data
NumPy	Numerical operations and array support
matplotlib	Builds charts and visual outputs
seaborn	Creates statistical plots and confusion-matrix heatmaps
scikit-learn	Encoding, train/test split, classifiers and evaluation metrics

12. REPRODUCIBILITY AND REFERENCES

12.1 Steps to Reproduce

- Open Google Colab at https://colab.research.google.com.
- Upload the project notebook named Netflix_Content_Strategy.ipynb.
- Run the notebook cells in order. Sign in to Kaggle if kagglehub requests authentication.
- If automatic dataset retrieval is unavailable, download netflix_titles.csv from the Kaggle dataset page and upload it to the Colab session.
- Execute the data preparation cells before running the visualizations and model-training cells.
- Review the five exploratory charts, model evaluation output, confusion matrices and Random Forest feature-importance chart.

Reproducibility note: the numerical metrics in this report are calculated from the confusion-matrix counts supplied with the project visuals. Re-running the notebook may produce different values if the dataset snapshot, preprocessing, library versions or notebook code changes.

12.2 Project Materials

File / Material	Description
Netflix_Content_Strategy.ipynb	Notebook containing data loading, cleaning, visualizations and model training
Netflix_Content_Strategy_Part1.pptx	Presentation covering the business problem, dataset and visualization plan
Supplied chart images	Five exploratory visualizations, two model confusion matrices and feature importance

12.3 References

- Kaggle. Netflix Movies and TV Shows dataset, published by shivamb. https://www.kaggle.com/datasets/shivamb/netflix-shows
- scikit-learn documentation. https://scikit-learn.org/
- pandas documentation. https://pandas.pydata.org/docs/
- NumPy documentation. https://numpy.org/doc/
- Matplotlib documentation. https://matplotlib.org/stable/
- seaborn documentation. https://seaborn.pydata.org/

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APPENDIX A - CLASS-WISE PERFORMANCE BREAKDOWN

The following class-wise scores and error counts are calculated from the confusion matrices in Figure 6.

Model	Audience group	Precision	Recall	F1-score	Support
Logistic Regression	Adults	0.560	0.847	0.674	802
Logistic Regression	Kids/Family	0.612	0.172	0.268	239
Logistic Regression	Teens	0.593	0.325	0.420	529
Random Forest	Adults	0.610	0.803	0.694	802
Random Forest	Kids/Family	0.615	0.435	0.510	239
Random Forest	Teens	0.595	0.389	0.471	529

A.1 Interpretation

Random Forest improves Kids/Family recall from 0.172 to 0.435 and F1-score from 0.268 to 0.510, indicating better identification of the smallest class, although more than half of its actual cases are still missed. Both models have higher recall than precision for Adults, while Teens remains difficult because many actual Teens titles are predicted as Adults. These class-wise results show why macro-averaged measures are more informative than accuracy alone for this imbalanced dataset.

A.2 Main Misclassification Patterns

The largest off-diagonal error for both models is actual Teens titles predicted as Adults. Random Forest reduces this error, but it remains substantial.

Actual audience group	Predicted group	Logistic Regression	Random Forest
Teens	Adults	349	298
Kids/Family	Adults	185	113
Adults	Teens	105	118

Random Forest reduces Teens-to-Adults errors from 349 to 298 and Kids/Family-to-Adults errors from 185 to 113, but Adults-to-Teens errors increase from 105 to 118. The mixed pattern shows why each class should be evaluated separately.

A.3 Evaluation Metric Definitions

Metric	Formula	Interpretation
Accuracy	Correct predictions / All predictions	Overall proportion classified correctly
Precision	TP / (TP + FP)	Share of predicted class members that are correct
Recall	TP / (TP + FN)	Share of actual class members identified
F1-score	2 x (Precision x Recall) / (Precision + Recall)	Balances precision and recall
Macro average	Mean of the three class-specific scores	Gives equal weight to each audience group